SoI Student Project Review

Smart Helmet for Rider Safety using IoT & Sensors

Team: META-HACKS | Lead: RAGHUL G

Members: NANDHA KUMAR C S, SANJAY T M, VISHNU VARDHAN A, SRIKANTH M

SoI Mentor: SRI RAM.R | SoI Vertical: EI|

Cohort/Batch: 2024 - 2028

Problem & End User Identification

Problem:

Road safety is a **critical challenge in India**, with millions of riders depending on two-wheelers. Many fatalities occur due to **alcohol consumption, drowsiness, and delayed medical response**. Current helmets only offer **passive protection**, not **active accident prevention or emergency support**.

End Users

- **Primary Users:** Two-wheeler riders who need enhanced safety.

- **Secondary Users:**
- Families of riders (who receive alerts).
- Traffic police (to detect violations).
- Hospitals & emergency services (for immediate rescue).
- Government authorities (to reduce road fatalities).

What situation triggers the problem?

The problem is triggered when riders operate two-wheelers under the influence of alcohol, leading to impaired judgment and accidents.

- **Alcohol consumption before or while driving** → causes accidents.
- **Rider fatigue or drowsiness** → loss of focus on the road.
- **Accidents in remote areas** → rider may be unconscious, delaying medical help.
- **No existing helmet feature** to automatically notify emergency contacts.
- **Rapid urbanization & high two-wheeler dependency** → increases accident risks.

Evidence of problem existence:

- **WHO Report (2023):** India accounts for **11% of global road deaths**.
- **MoRTH Data (2022):** Over **1.5 lakh deaths/year** due to road accidents.
- **Alcohol-related accidents:** Account for **16% of road crashes in India**.
- **Surveys:** Riders and families want **proactive safety solutions**.
- **Research Gap:** Most helmets only provide *impact resistance*; **IoT-enabled helmets are rare in adoption**.

Success metric(s):

- **Accident Prevention:** Reduction in drunk/fatigue-related crashes.
- **Emergency Response Time:** Immediate SMS/GSM alerts sent to families or hospitals.
- **System Reliability:** IoT-based monitoring with low failure rates.
- **User Acceptance:** Comfortable, affordable, and easy-to-use helmet.
- **Scalability:** Can be adopted widely in urban & rural regions.
- **Sustainability:** Supports *Digital India* and *Smart City* initiatives.

Problem Description

A **traditional helmet** only protects during **impact**, but it does not prevent the cause of accidents or help after a crash.

- The **Smart Helmet** introduces active safety features:
- **Alcohol detection sensor:** Prevents drunk driving.
- **Fatigue sensor (eye blink/heart rate):** Alerts rider if drowsy.
- **Crash detection (accelerometer/gyroscope):** Detects accidents instantly.
- **GSM/GPS module:** Sends emergency location & rider status to family/hospitals.
- **IoT connectivity:** Enables real-time monitoring and data analysis.
- Thus, the Smart Helmet is **not just protection gear, but a life-saving system**.

Feature / Criteria	Product A	Product B	Product C	Our Approach
Target User	General two-wheeler riders	Riders needing call/music	Premium riders	All two-wheeler riders
Core Function	Head protection only	Communication & music	Basic crash alert	Alcohol + Fatigue + Crash detection + IoT alerts
Accuracy/Performance	Low	Medium	Medium-high	High
Cost	₹800–2000	₹3000–6000	₹4000–7000	₹4000–8000
Deployment	Widely available	Limited markets	Imported, rare	Local design, scalable in India
Data/Privacy	None	Call/music data only	Limited IoT	Secure alerts via GSM/IoT
Known Limitations	No smart features	Distraction risk	not locally available	Needs GSM coverage, early prototype

Our Unique Selling Proposition (USP)

- **Value Proposition:** Smart helmet that **prevents accidents** and ensures **instant emergency alerts**.
- **Key Differentiators:** Alcohol detection • Fatigue monitoring • Crash alerts via GSM/GPS • IoT real-time monitoring • Affordable design.
- **Why Now:** Rising road accidents • Govt. push for smart safety • Low-cost IoT components available.
- **Evidence:** WHO & MoRTH reports • Pilot studies show rider acceptance • Safety demand in smart cities.
- **Feasibility:** Uses ESP32, GSM, MQ3, MPU6050 • Proven IoT tech • Low integration risk.

Project Plan & Timeline

Phase	Start–End	Milestones/Deliverables	Evidence of Validation
Problem & Customer Validation	SEP 2025	Accident statistics analysis, user interviews with riders, survey on helmet safety needs	WHO/MoRTH stats, survey results
Ideation (multiple concepts)	SEP 2025	Brainstormed multiple helmet concepts (alcohol detection, fatigue monitoring, crash alert, IoT integration)	Concept notes, mentor review
Proof of Concept (PoC)	OCT 2025	Breadboard setup with MQ3 alcohol sensor, MPU6050 accelerometer, GSM module connected to ESP32	Lab tests, SMS alerts
Prototype	JAN 2026	Integrated system inside helmet with sensors + ESP32 + GSM	Demo video, test results
Minimum Viable Product (MVP)	MAR 2026	Working smart helmet with all features (alcohol, fatigue, crash alert, IoT dashboard)	Field trials, user feedback
Final Product / Solution	MAY 2026	Fully functional helmet with ergonomic design, IoT monitoring, emergency alert integration	Final demo, report, validation

References

- WHO Global Road Safety Report, 2023
- MoRTH Road Accident Report, Govt. of India, 2022
- IEEE research papers on IoT-based smart helmets.
- Studies on alcohol and fatigue detection in transportation.
- IoT & embedded systems safety applications.